

Predicting IPR Patent Cancellation at Petition Filing

Final Project — Machine Learning · Esade S6 · May 2026 · Glossary and citations in Appendix A.

1. Problem Definition

An **Inter Partes Review** (IPR^[1]) is a USPTO administrative trial in which an accused infringer petitions to invalidate one or more claims of an issued patent, typically as a faster and cheaper alternative to district-court litigation. IPRs were created by the America Invents Act of 2011^[2] and are codified at 35 U.S.C. §§ 311–319. Each case typically resolves over 12–24 months in one of three ways: a **Final Written Decision** (FWD^[3]) cancelling all challenged claims, an FWD where some claims survive, or non-merits termination (settlement, institution denial, or discretionary denial under *NHK-Fintiv*^[4]).

Decision supported. At the moment of petition filing, both sides face a binary, high-stakes call — proceed, settle, or wait. The same call is currently made by attorneys reading the petition cold, which is slow, judgment-driven and expensive. A model-produced probability of cancellation — even if only modestly informative — supports this triage, *provided* its calibration profile is reported alongside it (see §5). **Target variable.** A binary cancelled in {0,1} label where 1 means the FWD held *all* challenged claims unpatentable (or the patent owner conceded by adverse judgment); 0 means any claim survived or the trial terminated non-merits.

Why supervised learning? Outcomes are observed and labelled by the USPTO Open Data Portal^[5], with rich pre- T_0 structured and textual evidence per filing — a textbook supervised setup, with the caveat that labels mature on a 12–24 month delay.

2. Dataset

We assembled a row-per-trial dataset by joining four structured surfaces and two PDF-text surfaces from the USPTO Open Data Portal: the proceedings, decisions, petitions, and patent-application records, plus the petition PDF text and the FWD PDF text (the latter used purely for label construction; the FWD verdict is not exposed in any structured field and lives only on the cover page of the FWD PDF). The raw cohort spans **15,030 IPR trials** filed between 2012-09-16 and 2026-01-05 (the post-AIA^[2] window). After dropping rows where the petition text was unrecoverable from the PDF (text shorter than 5,000 characters) or where the patent file wrapper had not been fetched, **14,822 trials** remain. We then apply a **600-day mature-label filter** (drop rows whose petition is more recent than ~20 months ago), since FWDs take a median of ~592 days to issue on this cohort and recent trials are mechanically dominated by fast-resolution outcomes (settlements, institution denials) all coded 0. The final modelling cohort is **13,596 trials**, with class balance shifting from 0.1748 to **0.1871** — confirmation that the mature filter was load-bearing.

3. Feature Engineering

The single inviolable constraint is **T_0 leakage discipline**: every feature must be observable strictly before the petition-filing date T_0 . The feature matrix is therefore *row-local* at T_0 ; any statistic that requires cross-row aggregation (frequencies, base rates) is deferred to the modelling preprocessor so it can refit per CV fold (§4). The transform layer (features/transforms.py) produces 35 raw features in seven families:

Family	n	Examples
Calendar	3	filing_year, filing_month, art_unit_group
Missingness pairs	4	prosecution_span_days(_missing), days_grant_to_petition(_missing)
Patent file-wrapper aggregates	11	n_events, n_assignments, n_distinct_assignees, n_parent_applications, n_pe / ex / aa / ad / iss / maint / other
Assignment recency / CPC breadth	4	days_since_last_assignment, no_recorded_assignment, n_cpc_codes, n_cpc_subclasses
Categorical — OHE	5	technology_center, cpc_section, ptab_era, entity_size, inventor_geo
Categorical — frequency	2	petitioner_real_party, owner_real_party
Petition-text (Tier A regex + length)	6	n_grounds, n_grounds_102/103, has_sotera_stipulation ^[6] , mentions_fintiv_factors ^[4] , petition_text_length

Two design choices deserve attention. First, **NaN-as-signal**: each nullable numeric feature gets a paired <name>_missing indicator set *before* imputation, so the missingness regime survives the median fill. Second, **regex precision discipline** for the petition-text family: features like has_sotera_stipulation and mentions_fintiv_factors are extracted from PDF-extracted text and audited via a three-pass workflow — true-hit contextual audit, false-classification broad-probe sweep for recall gaps, and a drop-or-broaden decision. Three candidate features (motivation_to_combine, general_plastic, aapa) were dropped because the false-sweep surfaced alternative phrasings that would have inflated false-positive risk; what survives is a small set of high-precision detectors rather than a stylometry of the petition. The audit was conducted on a 124-trial integration cohort stratified by year.

4. Modelling Pipeline

Preprocessor. The 35 raw features pass through a ColumnTransformer with four branches: (i) a six-fold PriorEncoder that emits a leakage-safe rolling cancellation rate and count for six identity / regime keys — petitioner, patent owner, petitioner×owner pair, technology center, patent number, and PTAB era; (ii) a FrequencyEncoder that maps the two open-vocabulary party fields to their training frequency (unseen categories map to 0); (iii) a OneHotEncoder over the closed-taxonomy categoricals with a `__missing__` sentinel level; and (iv) a median imputer on the remaining numeric columns. After expansion, the design matrix has roughly 82 columns. The preprocessor refits inside every CV fold so no statistic crosses a fold boundary.

The PriorEncoder is the load-bearing feature engineer. For each row at T_0 and each group key (e.g., the petitioner), it computes a rolling cancellation rate over training rows whose *label* resolved strictly before T_0 . The ordering by *label_resolution_date* — the FWD-issue date or, failing that, the termination date — rather than by *petition_filing_date* is deliberate: a co-filed trial whose label is not yet observable at T_0 cannot legitimately contribute to the prior. **Cold-start fallback:** for unseen groups (e.g., a first-time petitioner) the encoder returns the rolling global rate at T_0 rather than zero (which would imply ‘has lost every prior case’) or the corpus mean (which would leak future base rates). The six prior keys reflect the empirical ‘repeat-player advantage’ pattern documented in the IPR-outcome literature^[7].

Estimator and hyperparameters. We use a single RandomForestClassifier^[8] as the production model. The choice is deliberate: on a tabular dataset with mixed numeric, categorical, and frequency-encoded features, a tuned random forest is competitive with gradient boosting; it is robust to monotone transforms, handles missing-indicator columns natively, and supports stable permutation- and impurity-based importance for post-hoc interpretation, which a domain-expert audience needs. Hyperparameters: *n_estimators*=500, *max_depth*=None, *max_features*=0.3, *min_samples_leaf*=5, *class_weight*='balanced', *random_state*=42. *min_samples_leaf*=5 damps the variance of the rare-party prior branches; *class_weight* partially compensates for the ~19% positive base rate.

Validation strategy. We use a *forward-walking* time-series cross-validation on the pre-cutoff slice and reserve a *deployment-honest held-out tail* after a fixed cutoff (2023-01-01). Random K-fold leaks future cancellation rates into past predictions via the PriorEncoder and would inflate AUC. The CV folds snap to date boundaries so no same-day group straddles the train/test cut. CV scoring covers ROC-AUC, average precision and F1; the held-out tail is scored once and never used for model selection.

Choice of evaluation metrics. The legal user does not want a label, they want $P(\text{cancelled})$ for an expected-value calculation, so we report three complementary dimensions of fit: *ranking* (ROC-AUC, average precision), *calibration* (a reliability diagram comparing predicted probabilities to observed frequencies), and *threshold metrics* (precision / recall / F1 on the rare positive class). Accuracy alone is misleading at a ~19% base rate — a constant-zero classifier achieves ~81% accuracy with zero discriminative value. Asymmetric misclassification costs are real but case-specific (a false negative makes a winnable trial settle; a false positive proceeds a losing trial), so we deliberately do not bake a threshold into the model and instead expose the raw probability score together with the reliability diagram so the downstream user can pick a threshold that matches their own cost matrix.

5. Results

Metric	Value	Interpretation
Train rows / Held-out rows	11,689 / 1,907	Cutoff 2023-01-01; tail 433 cancelled
CV ROC-AUC (5-fold, mean $\pm$ std)	0.546 $\pm$ 0.034	Modest but above-chance signal on the pre-cutoff slice
Held-out ROC-AUC	0.576	Real but limited ranking power at T_0
Held-out Accuracy	0.775	Trails the 0.813 majority baseline by ~3.8pp; misleading
Held-out Precision (class 1)	1.000	Of trials flagged 1 at the 0.5 cutoff, ~100% truly cancel
Held-out Recall (class 1)	0.009	Catches only ~1% of true cancellations at default 0.5 cutoff
Held-out F1 (class 1)	0.018	Conservative on the rare class
Calibration	see Fig. 1	Mid-range bins over-confident; tail bins thin

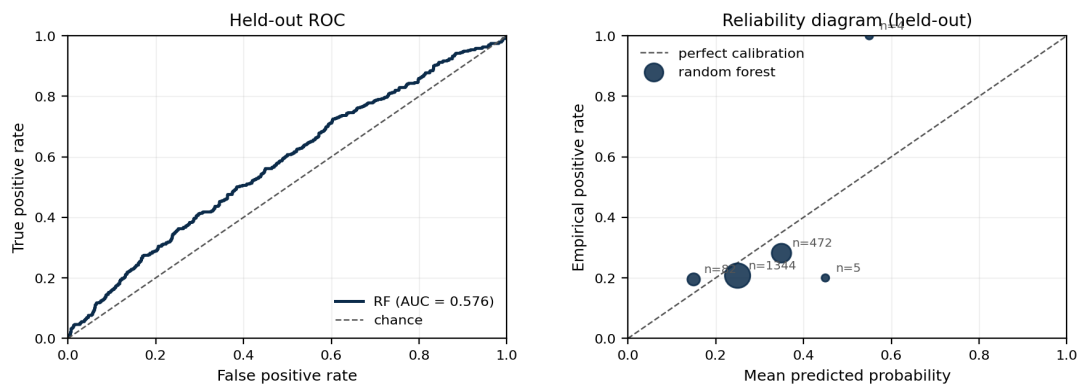


Figure 1. Held-out ROC curve (left) and reliability diagram (right). $AUC = 0.576$ confirms a modest ranking signal. The reliability scatter sits below the diagonal across the densely populated mid-range — the worst bin in support-weighted terms is $[0.2, 0.3)$ with $n = 1344$ trials, mean predicted 0.26 but only 0.21 actually cancelled. Marker area is proportional to bin support.

Three findings, three metric dimensions. First, ranking signal is *real but modest*: a held-out ROC-AUC of 0.58 on a held-out tail unseen by CV says the class boundary is genuinely hard at T_0 , which is consistent with the IPR-outcome literature finding that most cancellation evidence accrues during the trial rather than at filing^[7]. Second, the model is *over-confident in the densely populated mid-range*: the reliability diagram (Figure 1, right) shows that trials predicted to cancel with probabilities of 30–50% are over-stated — the observed cancellation frequency in those bins sits visibly below the diagonal. We deliberately do *not* fit a post-hoc probability calibrator on top of this random forest. A post-hoc calibration step is monotonic and so cannot improve ROC-AUC; an empirical check on this cohort showed that carving off even a small chronological slice of the most-recent training rows to fit a calibrator costs a meaningful chunk of held-out AUC, because those rows are exactly the ones closest in distribution to the held-out tail. That cost is the same regime-shift effect we document under *Distributional shift* in §6, and at our signal level it dominates the calibration benefit.

Reporting the reliability diagram alongside the raw probability is therefore the more honest move — a downstream user reading Figure 1 sees the mid-range over-confidence directly, rather than buried under a calibrator that would mask it. Third, the threshold metrics show the cost of class imbalance: at the default 0.5 cutoff the model recovers only ~1% of true positives. `class_weight='balanced'` already shifts the estimator toward the minority class; pushing further is a threshold choice the downstream user should make against their own asymmetric cost matrix.

6. Critical Reflection

Construct validity (label).

cancelled measures legal cancellation, not ‘did the petitioner achieve their objective.’ A negotiated settlement that buys a favorable license is coded 0, even though the petitioner may regard the outcome as a win. A partial FWD that knocks out most — but not all — challenged claims is also coded 0, even though it may have substantially weakened the patent's enforceability against the petitioner. The label is unambiguous in the regulatory record but lossy as a proxy for business outcome — a different deployment might require relabelling.

Construct validity (text features).

Regex over PDF-extracted text measures firm *style*, not legal *substance*, until the false-sweep audit catches the gap. Three candidate features were dropped on this basis. The five surviving Tier-A detectors survive manual audit on a 124-trial integration cohort, but the underlying risk — that we are measuring ‘do counsel write the canonical phrase?’ rather than ‘is the argument made?’ — never goes to zero. A future iteration would benchmark these features against an LLM judge over the same cohort.

Distributional shift.

The `ptab_era` feature is structural: a USPTO Director change literally redefines $P(Y | X)$ through guidance memos and discretionary-denial policy^[9]. The latest era (`trump_2_post_vidal`, January 2025 onward) is severely under-represented in training, yet 100% of deployment inputs would lie in it. Forward-walking CV partly mitigates by scoring on later folds, but a regime-aware retraining cadence — at minimum on every Director memo — is an operational requirement, not a nice-to-have.

Fairness.

Five of the six PriorEncoder keys are identity-based (petitioner, owner, pair, technology center, patent number); only `ptab_era` is regime-shaped. The encoder already exposes the `<group>_prior_count` alongside the rate, so the estimator can in principle discount low-history rows; in practice, first-time and pro-se petitioners still cold-start to the rolling global mean and the model encodes the empirical ‘repeat-player advantage’^[7] rather than questioning it. Predictions therefore reinforce a known structural inequity in the IPR system and risk chilling legitimate first-time challenges. Mitigations that go beyond the existing feature set would include: stratified reliability audits by prior-count decile (to test whether the over-confidence on cold-start rows is genuinely larger than the rest of the reliability noise); surfacing a prediction *uncertainty band* derived from the per-tree variance of the forest, so downstream users see ‘ 0.40 ± 0.05 (200 prior trials)’ vs. ‘ 0.30 ± 0.18 (cold start)’; and offering an opt-in model variant with the petitioner-identity branch ablated for policy or pro-bono users where the repeat-player signal is normatively unwanted.

Information ceiling at T_0 .

The hardest limit is informational rather than methodological. Leakage discipline excludes by construction every signal that accrues after filing — the institution decision, claim-construction order, deposition-stage motions, expert reports — all of which the IPR-outcome literature identifies as the dominant predictors of cancellation^[7]. A 0.58 held-out AUC at T_0 is therefore close to a ceiling rather than evidence of an under-tuned model; meaningful gains likely require either (a) a staged-prediction setup with a refresh after institution, or (b) substantially richer petition-text representations (an LLM-derived legal argument quality score). Both are deferred to future work.

Honest deployment posture.

We treat the model as *decision support*, not autonomous decisioning. What the pipeline actually outputs today is a single raw probability score per trial — uncalibrated by design (see §5: a post-hoc calibrator would cost more held-out ranking power than it would buy back at this signal level) and without an uncertainty band. The reliability diagram (Figure 1, right) is shipped alongside the score so the mid-range over-confidence is visible to the user rather than hidden. The per-tree-variance uncertainty band described under *Fairness* is flagged as next-iteration work rather than shipped, and the report does not claim it as a current feature. Attorneys, not the model, own the call. The metric we would track in production is not held-out AUC but the drift of the reliability curve on each subsequent year's FWDs, with an automatic retrigger to retrain on regime change.

Reproducible code in repo. Entry points: `drivers/run_features.py`, `final_submission/run_training_for_report.py`. Standalone notebook: `final_submission/notebooks/01_getting_started.ipynb`.

Appendix A. Glossary and Citations

Numbered superscripts in the main text refer to the entries below. Glossary entries cover patent-law and ML terminology used in the report; citations point to primary sources or representative scholarship.

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- [1] **IPR — Inter Partes Review.** A USPTO administrative trial established by the AIA^[2]. Petitioners challenge claim validity under 35 U.S.C. §§ 102 (novelty) and 103 (non-obviousness); the trial is heard by a three-judge panel of the Patent Trial and Appeal Board (PTAB). See 35 U.S.C. §§ 311–319.
 - [2] **AIA — Leahy-Smith America Invents Act,** Pub. L. No. 112-29, 125 Stat. 284 (Sept. 16, 2011). Created the IPR, post-grant review, and covered-business-method review proceedings, and introduced the first-inventor-to-file system. The 2012-09-16 effective date is why our cohort starts on that calendar day.
 - [3] **FWD — Final Written Decision.** The merits judgment issued by the PTAB at the conclusion of an instituted IPR, addressing the patentability of each challenged claim. Required by 35 U.S.C. § 318(a) within one year of institution (extendable by six months for good cause).
 - [4] **NHK-Fintiv discretionary denial.** Doctrine under which the PTAB may deny institution of an otherwise meritorious IPR petition when parallel district-court litigation is advanced. Established by *NHK Spring Co. v. Intrix-Plex Techs., Inc.*, IPR2018-00752 (PTAB Sept. 12, 2018) (designated precedential), and refined by *Apple Inc. v. Fintiv, Inc.*, IPR2020-00019 (PTAB Mar. 20, 2020) (designated precedential), which articulated the six-factor balancing test now known as the ‘Fintiv factors.’
 - [5] **USPTO Open Data Portal (ODP).** Public REST API surfacing PTAB proceedings, decisions, petitions, and patent-application metadata. <https://data.uspto.gov/>.
 - [6] **Sotera stipulation.** A petitioner’s commitment, before institution, not to pursue in parallel district-court litigation the same invalidity grounds raised in the IPR. Originated in *Sotera Wireless, Inc. v. Masimo Corp.*, IPR2020-01019 (PTAB Dec. 1, 2020) (designated precedential). A Sotera stipulation strongly weighs against discretionary denial under the *Fintiv*^[4] framework.
 - [7] Empirical IPR-outcome literature documenting repeat-player advantages and identifying institution-stage and post-institution signals as dominant predictors of cancellation. See, e.g., S. Vishnubhakat, A. Rai & J. Kesan, *Strategic Decision Making in Dual PTAB and District Court Proceedings*, 31 Berkeley Tech. L.J. 45 (2016); B. Love & S. Ambwani, *Inter Partes Review: An Early Look at the Numbers*, 81 U. Chi. L. Rev. Dialogue 93 (2014).
 - [8] **Random forest.** L. Breiman, *Random Forests*, 45 Mach. Learn. 5 (2001). Implementation: `sklearn.ensemble.RandomForestClassifier` (Pedregosa et al., *Scikit-learn: Machine Learning in Python*, 12 J. Mach. Learn. Res. 2825 (2011)).
 - [9] **Director memoranda.** The USPTO Director periodically issues binding guidance memoranda that reweight or temporarily suspend the *Fintiv*^[4] factors. Notable examples in the cohort window: Director Vidal’s June 21, 2022 memo (declining discretionary denial when a Sotera stipulation is filed) and the January 20, 2025 administration transition. Each memo creates a regime boundary that the `ptab_era` feature is designed to capture.
 - [10] **T_0 .** Internal notation in this report and codebase for the `petition_filing_date` — the moment of prediction. Every feature must be observable strictly before T_0 ; the ‘leakage discipline’ in the feature pipeline enforces this row-by-row.
 - [11] **Mature-label filter.** Drops trials whose petition is more recent than 600 days ago, since FWDs take a median of ~592 days to issue on this cohort. Without this filter, recent trials (which haven’t yet had time to reach FWD) are mechanically dominated by fast-resolution non-merits outcomes (settlements, institution denials), all coded 0 — inflating held-out accuracy without genuine signal.
 - [12] **Forward-walking time-series CV.** Standard practice for temporally ordered data; see, e.g., R. Hyndman & G. Athanasopoulos, *Forecasting: Principles and Practice* (3d ed. 2021), § 5.10.
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